

TELCONG NETPLAN

Network Planning & Regulatory Affairs

Plot 274 Adetokunbo Ademola Crescent, Wuse II, Abuja, FCT
tel: +234 9 461 7000 | fax: +234 9 461 7099 | www.telcong-netplan.example

5G CBD Coverage Gap Analysis

Site density justification for the Abuja CBD 5G Densification Pilot

Authors	Sheila Stokes + Emeka Adeniyi (Capacity Planning)
Date	2026-09-18
Scope	Abuja CBD: Wuse, Maitama, Garki, Asokoro, CBD proper (28 km²)
Methodology	Drive test 2026-Q2 + Ericsson 9999 propagation model
Use	Form NCC-7B Section D, Annex D-2

Executive summary

The Abuja CBD currently exhibits a measured coverage and capacity gap relative to comparable urban CBDs in Nigeria, justifying the proposed 24-site 5G NR densification pilot. This document presents the measurement evidence, the gap quantification, and the projected post-pilot performance.

1. Coverage area definition

The Abuja CBD is defined as the contiguous area bounded by the inner ring road, comprising Wuse, Maitama, Garki, Asokoro, and the Central Business District proper. The area is approximately 28 km² with daytime population of 95,000 workers and residential population of 38,000. Major landmarks include the National Assembly, the Federal Secretariat, the Central Bank of Nigeria, the major bank headquarters (GTBank, Zenith, UBA, Access), the Sheraton and Hilton hotels, and the foreign embassy district.

2. 2026-Q2 drive test results (baseline)

TelcoNG NetPlan conducted a drive test campaign of the CBD in April-May 2026 (2026-Q2). The campaign measured downlink throughput, latency, RSRP, RSRQ, and SINR across the existing 4G LTE-A grid operated by the four major incumbents. Results below.

Metric	Median	p95 worst	Industry target	Gap to target
DL throughput (Mbps)	22	4	≥100	78% below
UL throughput (Mbps)	8	1.5	≥40	80% below
DL latency (ms)	64	180	≤20	3.2x above
Voice setup time (ms)	1,250	2,800	≤800	1.6x above

Metric	Median	p95 worst	Industry target	Gap to target
RSRP (dBm)	-92	-118	≥-95	Pass median; fail p95
RSRQ (dB)	-12	-22	≥-13	Pass median; fail p95
SINR (dB)	8	-2	≥6	Pass median; fail p95
Call drop rate (%)	0.4	2.1	≤0.5	Pass median; fail p95
Handover success rate (%)	98.2	92.4	≥99.0	0.8% below median
Service availability (%)	99.6	97.8	≥99.95	0.35% below median

3. Identified hotspots

The campaign identified 12 specific hotspots where p95 performance is worst, concentrated in office-tower clusters and high-density mixed-use areas:

#	Hotspot	Area (m²)	Daytime pop	p95 DL (Mbps)	Primary cause
1	National Assembly complex	62,000	8,500	1.2	Building shadowing + high density
2	Federal Secretariat block A-D	120,000	12,000	1.8	High density + 7-floor towers
3	CBN HQ + Adetokunbo Ademola corridor	45,000	6,200	2.4	Building shadowing
4	GTBank HQ + GRA boundary	31,000	4,800	3.1	High density
5	Sheraton Abuja + Aguiyi Ironsi corridor	28,000	3,400	3.4	Mixed-use density
6	Central Bank corner + Plot 274 cluster	18,000	2,900	2.1	Building shadowing
7	Wuse Market and surrounds (weekend peak)	85,000	22,000	0.8	Extreme density (weekend)
8	Garki II business district	44,000	5,800	2.6	Mixed-use density
9	Maitama embassy cluster	39,000	1,200	2.9	Building shadowing
10	Asokoro residential ring	62,000	3,500	3.6	Lower density but coverage gap
11	Hilton Abuja + Independence Avenue	26,000	3,100	2.8	Mixed-use density
12	Eagle Square + ECOWAS secretariat	34,000	1,800	3.2	Building shadowing

4. Proposed 24-site densification

The proposed pilot deploys 24 5G NR sites in a densified grid covering the 28 km² CBD footprint. Inter-site distance is approximately 600 metres, comparable to international 5G mid-band deployments in Tokyo, Seoul, and London. Each site is configured with 3 sectors and 1 BBU controlling 4 radio units per sector. The total radio inventory is 14 RUs initially provisioned with growth capacity to 32 RUs (24 sites × 3 sectors × 4 RUs = 288 RU slots, of which 14 are active at launch).

Sub-area	Sites in pilot	Inter-site (m)	Pop served (daytime)	Coverage (km ²)
Wuse	6	580	23,000	6.2
Maitama	4	650	12,000	5.8
Garki	5	590	18,000	5.4
Asokoro	4	720	8,500	5.2
CBD proper (Central Area)	5	560	33,500	5.4
TOTAL	24	Avg 620	95,000	28.0

5. Predicted post-pilot performance

Coverage simulation using TelcoNG's calibrated Ericsson 9999 propagation model predicts the following post-pilot performance across the 28 km² CBD footprint:

Metric	Baseline (4G)	Post-pilot (5G NR)	Improvement
Median DL throughput (Mbps)	22	380	17.3×
p95 worst DL throughput (Mbps)	4	85	21.3×
Median UL throughput (Mbps)	8	90	11.3×
Median DL latency (ms)	64	18	3.6× better
p95 worst DL latency (ms)	180	42	4.3× better
Service availability (%)	99.6	99.95	+0.35 pp
Cell-edge SINR (dB, p10)	-2	4	+6 dB
Capacity per km ² (Gbps)	0.8	12.4	15.5×

6. Conclusion

The measured coverage gap in the Abuja CBD is substantial, particularly in the p95 worst-case performance that defines the user experience in the most demanding locations. The proposed 24-site 5G NR densification, operating in the 100 MHz contiguous block at 3500-3600 MHz, addresses the gap with a 17× throughput improvement and a 3.6× latency improvement at the median, with even larger improvements at the p95. The site density (24 sites over 28 km², approximately 600 m inter-site) is consistent with international 5G mid-band best practice. The pilot is technically justified.